

Rahul Mate

Senior Machine Learning Engineer

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Experienced Senior Machine Learning Engineer with over 5 years of focused expertise in developing, fine-tuning, and deploying large-scale language models, ASR systems, and AI-powered applications. Skilled in leveraging multi-GPU environments for optimized fine-tuning, and crafting real-time, high-throughput NLP services. Known for driving cost efficiencies, enhancing model performance, and creating tailored solutions that meet unique client demands across industries.

Technical Skills

- **Languages:** Python, Bash
- **Frameworks:** DeepSpeed, PyTorch, TensorRT, FastAPI, ONNX
- **Libraries:** spaCy, NLTK, Transformers, TensorFlow
- **Tools:** Kubernetes, Docker, Hazelcast, GCP
- **Methods:** Quantization, Fine-tuning, Large Language Models (LLMs), Automatic Speech Recognition (ASR), Speaker Diarization

Experience

Senior Machine Learning Engineer

syml.ai, pune, Maharashtra

March 2019 - Present

Async ASR Model:

- Created an in-house ASR model, exploring and conducting POCs with various implementation methods.
- Led end-to-end development on TensorRT engines to build and deploy the model.
- Evaluated model performance and accuracy, followed by implementing advanced features.
- Developed an evaluation pipeline to benchmark model outputs against other vendors.
- Conducted iterative experiments to enhance model performance and quality.
- With this Project we reduce the ASR production cost by 2x and improved overall accuracy by 30 percent.

Speaker Diarization:

- Explored multiple speaker diarization methods and models, assessing them by latency and accuracy.
- Built a pipeline to evaluate the performance of diarization models.
- Contributed to optimizing the deployment process for diarization models.

Optimized Language Model Fine-Tuning

- Fine-tuned large models (Falcon, Phi3, Llama) using DeepSpeed across multi-GPU/multi-node setups, reducing memory usage by 50% and doubling training speed with Deepspeed.

Large Language Model Inference Optimization

- Developed and deployed a high-throughput FastAPI service for LLMs, using quantization techniques (4-bit, 8-bit, GPTQ) and TGI to optimize throughput and reduce memory overhead.

Language Model Evaluation

- Integrated custom metrics and datasets into Stanford CRFM's HELM framework and lm-evaluation-harness, enhancing evaluation transparency and model performance.
- Evaluated language models using MT-bench, improving their overall effectiveness.

Answering Machine Detection Model

- Developed a beep detection model from scratch and used Tracker for a sales call use case.
- Implemented various machine learning techniques to enhance model performance and enable real-time processing.

Analytics API

- Analyzed customer requirements for an analytics API to provide speaker-level information and metadata.
- Implemented a service that delivers speaker-specific insights, serving as a metric for sales training and assistance.

Sales Call Follow-Up and Action Item Detection System

- Enhanced a classification model to detect follow-up actions in sales call transcripts, improving detection accuracy.
- Optimized deployment by converting the model from PyTorch to TensorRT, reducing latency and hardware usage by 50%.
- Used GPU sharing Node Deployment to optimize GPU memory usage and increase concurrency of the model.
- Employed spaCy and NLTK to reduce false positives and incorporated reverse coreference for accurate extraction of appointment details.

Topic Detection Model

- Optimized an in-house topic detection model, focusing on both accuracy and latency improvements.
- Implemented and tested the model to detect topics in real-time scenarios.

Sentiment Model

- Led the end-to-end development, testing, and deployment of a sentiment analysis model.
- Integrated the sentiment model with the topic detection model to provide topic-level sentiment insights.

Big Query Pipeline

- Developed a Python pipeline to update BigQuery tables for customer dashboards.
- Set up a cron job using CircleCI to refresh data at specified intervals.

Education

Bachelor of Engineering in Electronics and Telecommunication Engineering
YCCE , nagpur

Aug 2017
